

BRISC2025 Segmentation Inference Report

26 May 2026

Goal. This report summarizes the currently trained BRISC2025 stages, with focus on the single-kernel radial learning results and updated inference visuals.

Method Overview. The workflow includes: (1) selected-kernel classification from a broad filter bank, (2) a combined-kernel baseline, and (3) radial single-kernel optimization (seed, learned, and monotone-constrained variants).

Structured and Combined Stages.

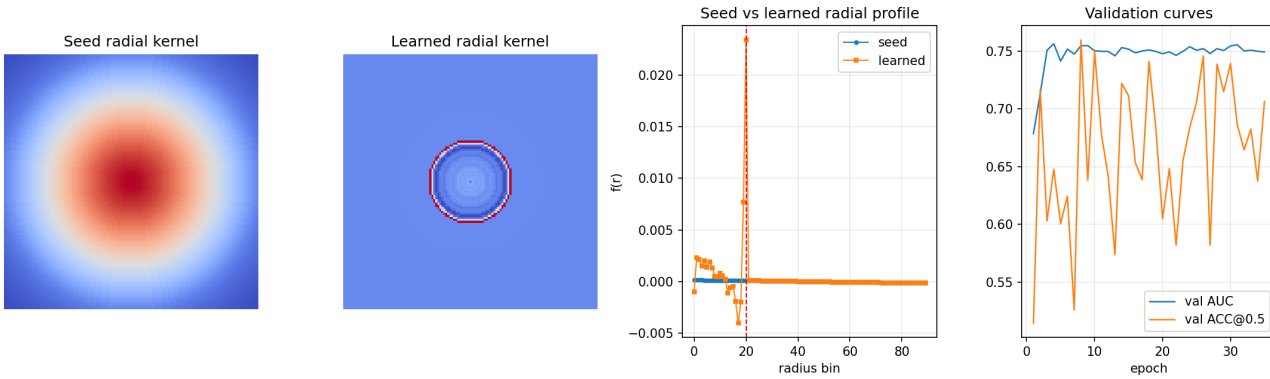
Model Stage	Val Acc	Val AUC	Test Acc	Test AUC
Selected-kernel classifier	0.7398	0.8366	0.7715	0.8321
Combined-kernel baseline	0.6764	0.6994	0.6225	0.7013

Single-Kernel Radial Stages. The values below use the validation-tuned threshold.

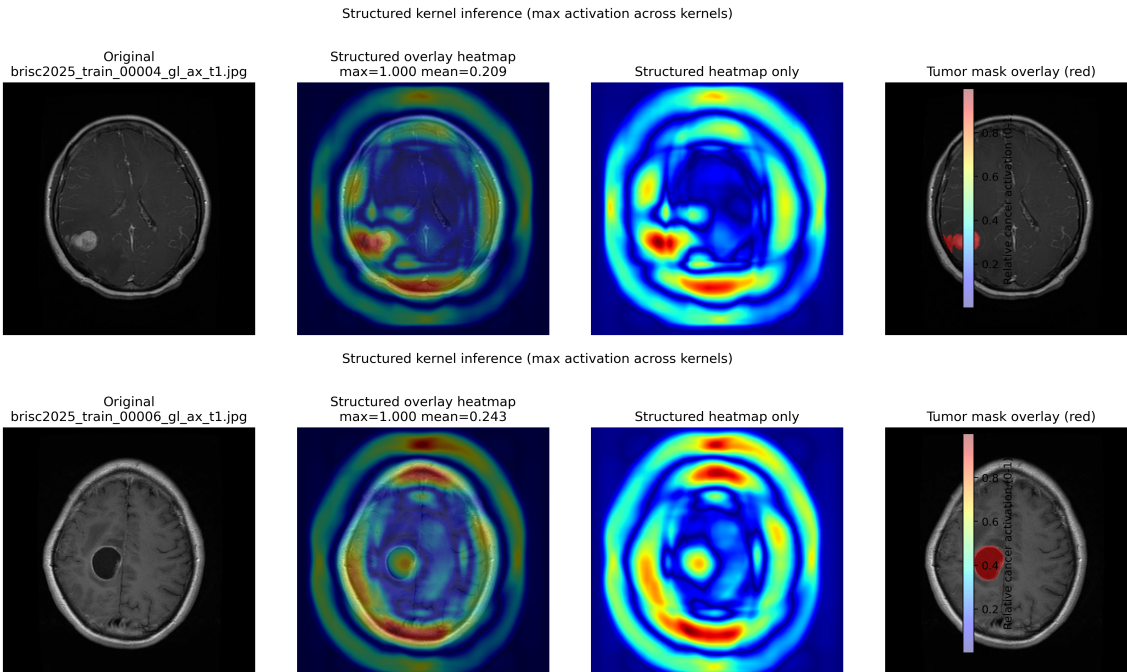
Radial Stage	Val Acc	Val AUC	Test Acc	Test AUC
Seed single-kernel radial	0.7285	0.7012	0.6405	0.6961
Learned single-kernel radial	0.7536	0.7563	0.6740	0.7303
Monotone-constrained radial	0.6953	0.6801	0.6962	0.6268

The learned single-kernel stage improves over the seed radial baseline in both validation and test AUC, while the monotone-constrained version gives a more constrained shape and slightly different accuracy tradeoff.

Single-Kernel Trained Figure.

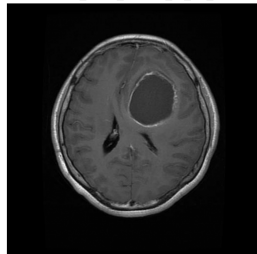


Structured-Kernel Heatmaps (Rows).

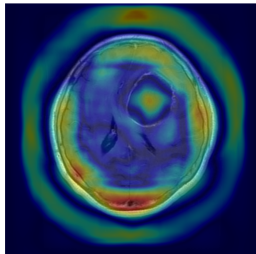


Structured kernel inference (max activation across kernels)

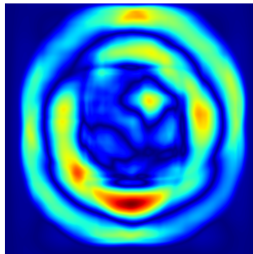
Original
brisc2025_train_00007_gl_ax_t1.jpg



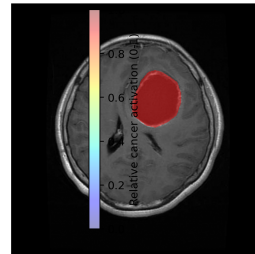
Structured overlay heatmap
max=1.000 mean=0.220



Structured heatmap only

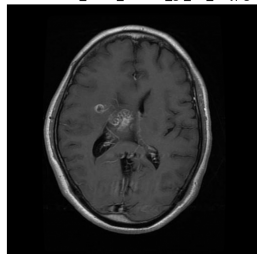


Tumor mask overlay (red)

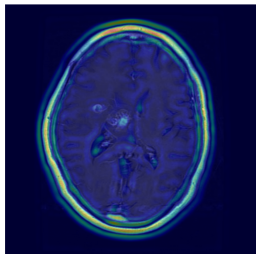


Structured kernel inference (max activation across kernels)

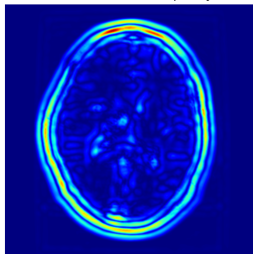
Original
brisc2025_train_00005_gl_ax_t1.jpg



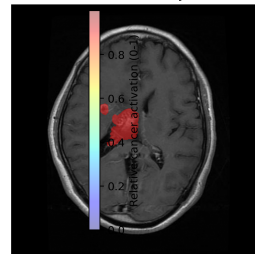
Structured overlay heatmap
max=1.000 mean=0.074



Structured heatmap only



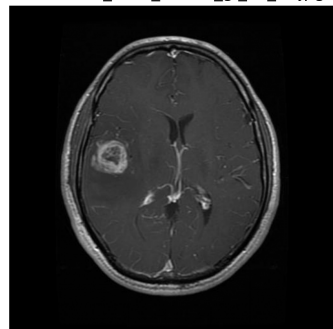
Tumor mask overlay (red)



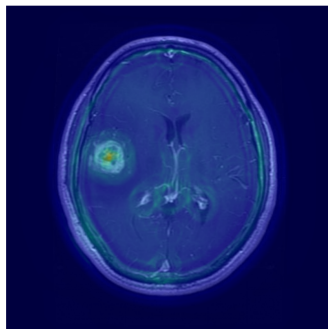
Learned Radial Heatmaps (Rows).

Learned radial kernel inference | Test AUC=0.730 Test ACC@0.5=0.649 Test ACC@thr_val=0.674

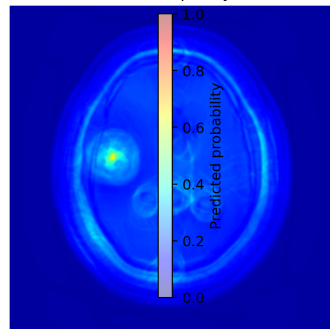
Original
brisc2025_train_00003_gl_ax_t1.jpg



Overlay heatmap
max=0.682 mean=0.100

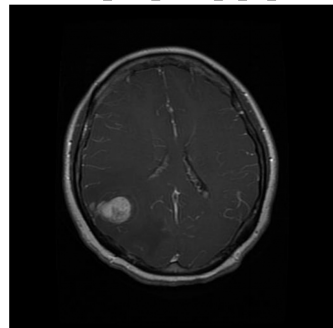


Heatmap only

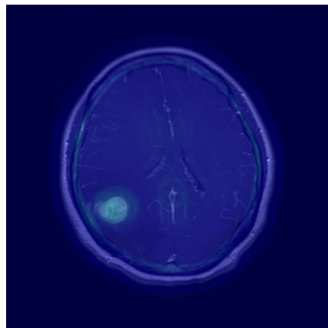


Learned radial kernel inference | Test AUC=0.730 Test ACC@0.5=0.649 Test ACC@thr_val=0.674

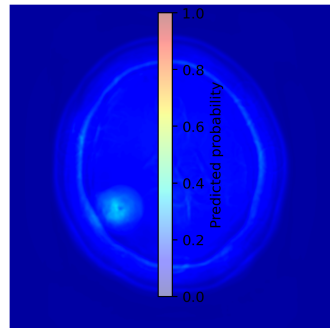
Original
brisc2025_train_00004_gl_ax_t1.jpg



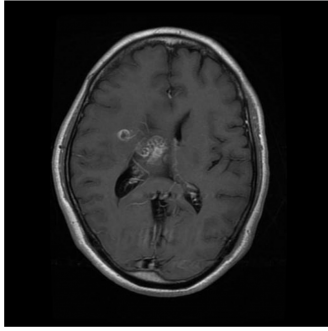
Overlay heatmap
max=0.350 mean=0.071



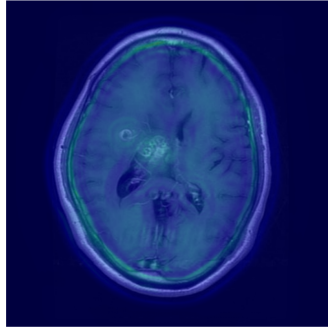
Heatmap only



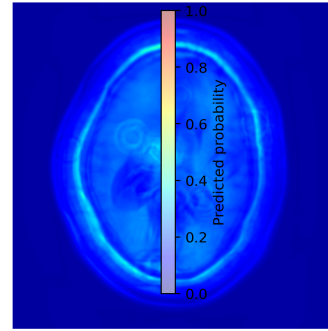
Original
brisc2025_train_00005_gl_ax_t1.jpg



Overlay heatmap
max=0.438 mean=0.109



Heatmap only



Conclusion. The selected-kernel model remains strongest for final predictive quality, and the updated single-kernel radial training adds an interpretable local-response view with clear learned heatmaps across multiple BRISC2025 cases.